

The cost of the desert is not a Slepian hole hole then sub-Nyquist, not a sum of holes

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Abstract

A pure Slepian estimate on the first gap $(\gamma_1 - 2\pi/L)_+$ underpredicts $-\ln c_L$ by a factor 2.5 at $\mu = 3$ and 5–6 at $\mu = 30$ –50. The factor moves with μ and is never 1. The additive law $aL(\gamma_1 - 2\pi/L)_+ + bL \sum (\text{gap} - 2\pi/L)_+$ with $a = 1.69$, $b = 0.82$ is a local fit, not the sampling constant of the set. The missing object is Beurling–Landau for a hole *followed by* a sub-Nyquist stretch — one set, not two terms.

1 Slepian on the hole alone

Take $f \in PW_{L/2}$, even. The prolate problem with frequency support in the desert $[0, \gamma_1]$ has parameter $c = \frac{L}{2}(\gamma_1 - 2\pi/L)_+$. The leakage out of the desert is

$$1 - \lambda_0(c) \sim 4\sqrt{\pi c} e^{-2c}.$$

If later zeros sampled the leakage at Nyquist and independently, one would have $-\ln c_L \approx 2c = -\ln(1 - \lambda_0)$ up to $\frac{1}{2} \ln c$. That is the “naive $a = 2$ ”.

Measured against that prediction:

name	μ	$2c$	$-\ln \text{Slep}$	$-\ln \lambda$	ratio
ζ	3	9.25	6.52	16.69	2.56
ζ	11	27.6	24.3	109.3	4.49
χ_5	30	16.3	13.3	68.1	5.11
χ_5	50	19.7	16.6	108.5	6.52
χ_{17}	11	2.66	0.56	4.89	8.8

The hole explains the floor *up to a factor*. The factor is 2.5 on the smallest window and climbs. It is not 1.69/2 either: writing $-\ln \lambda = a_{\text{eff}} L(\gamma_1 - 2\pi/L)_+$ gives $a_{\text{eff}} = 1.81$ at $\mu = 3$ (ζ) and 4.2–5.5 at χ_5 , $\mu = 30$ –50. So $a \in [1.7, 2.1]$ is the ζ fit on $\mu = 3$ –16, not a constant of the hole.

2 Why adding gaps does not repair it

The first zeros after γ_1 are themselves wider than $2\pi/L$. Treating each excess as a new Slepian hole and summing the exponents assumes that a test function can hide in one gap without seeing the others. The function that realises the floor is (to leading exponential) the first prolate of the desert: its Fourier tail on $[\gamma_1, \gamma_*]$ is fixed, one-sided, and already correlated with those zeros. The extra cost is the sampling constant of

$$\Lambda_{\text{early}} = \{\gamma_k : \gamma_k \leq \gamma_*\}$$

as a single non-uniform set for $PW_{L/2}$, not $\text{cost}(\text{desert}) + \sum \text{cost}(\text{gap}_k)$.

That is why b in the additive fit is 0.82 rather than 1, and why s_{pred} drifts once μ leaves the window where (a, b) were pinned.

3 What Beurling–Landau ask for

A discrete set $\Lambda \subset \mathbb{R}$ is sampling for PW_Ω if and only if its exterior density satisfies $D^-(\Lambda) \geq \Omega/\pi$ (Beurling) and a uniform separation holds. Here $\Omega = L/2$, so the Nyquist density is $L/(2\pi)$, spacing $2\pi/L$. On $[0, \gamma_1]$ one has $D^- = 0$. On $[\gamma_1, \gamma_*]$ the density climbs from 0 to Nyquist through the measured gaps. The sampling constant of that union is a classical object (Landau’s completeness radius, Slepian on a multi-band frequency set). It is not computed by adding two Slepian exponents.

A computable proxy, not used as a theorem: the first prolate ψ_0 of the desert, then $\sum_k |\hat{\psi}_0(\gamma_k)|^2$. That sum is the quadratic form of Q_z on one vector. It reproduces the leading exponential only if ψ_0 is already close to the ground state of Q ; MUSIC says it is not — the ground state is a Slepian of the *whole* early set.

Status

Diagnosed. The hole gives the right exponential *shape* and the wrong constant, by a μ -dependent factor 2.5–6. The additive (a, b) law is a chart, not Beurling. The missing analysis is the sampling constant of a hole plus a sub-Nyquist prefix, as one set. Not RH.